

Hypoplactic Classes Are Not Counted by Plane Partitions: A Disproof of Conjecture 00000003842

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Abstract

The source asserts that the number of hypoplactic classes of words of length at most ℓ over an n -letter alphabet equals the number of plane partitions in a $2 \times n \times \ell$ box. This is false. Over two letters there are 7 classes of length at most 2 and 4 of length exactly 2, but 20 plane partitions in the $2 \times 2 \times 2$ box. The identity also fails for every $\ell \geq 1$ when $n = 1$ and for every $n \geq 1$ when $\ell = 1$. The hypoplactic monoid, its classes, plane partitions, and the negation of the identity are formalised in Lean. An earlier submission had the right example but no formal content; its error is described below.

Definitions and reading of the source

Let $A = \{1 < 2 < \dots < n\}$ and let A^* be the free monoid of words. The *plactic congruence* is generated by the Knuth relations

$$acb \equiv cab \quad (a \leq b < c), \quad bac \equiv bca \quad (a < b \leq c).$$

The *hypoplactic congruence* $\equiv$ is generated by the Knuth relations together with the quartic relations

$$cadb \equiv acbd \quad (a \leq b < c \leq d), \quad bdac \equiv dbca \quad (a < b \leq c < d).$$

The hypoplactic monoid is $A^*/\equiv$; it is a quotient of the plactic monoid (Krob–Thibon, Novelli). All defining relations preserve length, so every class has a length.

A *plane partition in an $a \times b \times c$ box* is an $a \times b$ array of integers in $\{0, \dots, c\}$ that is weakly decreasing along every row and every column. Their number $\text{PP}(a, b, c)$ is symmetric in a, b, c , so the order of the three sides in the source does not matter.

The English text counts the classes “of length at most ℓ ”. The Chinese text counts the classes “of length ℓ ”. We treat both. Write $h(n, \ell)$ for the number of classes of words of length exactly ℓ , and $H(n, \ell) = \sum_{m \leq \ell} h(n, m)$ for the number of classes of words of length at most ℓ , the empty word included. The source claims $H(n, \ell) = \text{PP}(2, n, \ell)$, or in the Chinese wording $h(n, \ell) = \text{PP}(2, n, \ell)$, for all n and ℓ .

The disproof

Lemma 1. *If $u \equiv v$ then u and v have the same length. If moreover $u \neq v$, then this length is at least 3.*

Proof. Let R be the relation on A^* defined by: $u R v$ if $u = v$, or u and v have the same length and that length is at least 3. It is an equivalence relation. It is compatible with products: if $u R v$ and $u' R v'$, then either $u = v$ and $u' = v'$, so $uu' = vv'$, or one of the two pairs consists of words of length at least 3, and then uu' and vv' have the same length, at least 3. So R is a congruence. Both sides of every defining relation have length 3 or 4, so R contains the defining relations. Hence R contains $\equiv$. $\square$

Theorem 2. (a) For $\ell \leq 2$, $h(n, \ell) = n^\ell$. Hence $H(n, 1) = n + 1$ and $H(n, 2) = n^2 + n + 1$.

(b) $h(1, \ell) = 1$ and $H(1, \ell) = \ell + 1$ for every ℓ .

(c) $\text{PP}(2, 1, \ell) \geq \ell + 2$ for $\ell \geq 1$, $\text{PP}(2, n, 1) \geq n + 2$ for $n \geq 1$, and $\text{PP}(2, 2, 2) = 20$.

Consequently

$$h(2, 2) = 4, \quad H(2, 2) = 7, \quad \text{PP}(2, 2, 2) = 20;$$

for every $\ell \geq 1$, $h(1, \ell) < H(1, \ell) < \text{PP}(2, 1, \ell)$; and for every $n \geq 1$, $h(n, 1) < H(n, 1) < \text{PP}(2, n, 1)$. Both readings of the conjectured identity are false.

Proof. (a) By Lemma 1 a word of length at most 2 is alone in its class. So the classes of length $\ell \leq 2$ correspond bijectively to the n^ℓ words of length ℓ . Classes of different lengths are different, which gives the values of H .

(b) Over one letter there is exactly one word of each length, and by Lemma 1 words of different lengths are inequivalent.

(c) In the $2 \times 1 \times \ell$ box the $\ell + 1$ constant arrays and the array with entries 1 above 0 are $\ell + 2$ different plane partitions when $\ell \geq 1$. In the $2 \times n \times 1$ box, take for $k = 0, \dots, n$ the array whose first row is $1^k 0^{n-k}$ and whose second row is zero, and the array with all entries 1; these are $n + 2$ different plane partitions when $n \geq 1$. A plane partition in the $2 \times 2 \times 2$ box is an array $\begin{pmatrix} x & y \\ z & w \end{pmatrix}$ with $2 \geq x \geq y \geq w \geq 0$ and $x \geq z \geq w$. For fixed x and w there are $(x - w + 1)^2$ choices of (y, z) , so the number is $1 + (4 + 1) + (9 + 4 + 1) = 20$.

The displayed consequences follow: $4 < 7 < 20$; $1 < \ell + 1 < \ell + 2$ for $\ell \geq 1$; and $n < n + 1 < n + 2$. $\square$

Remark. The exact values are $\text{PP}(2, 1, \ell) = \binom{\ell+2}{2}$ and $\text{PP}(2, n, 1) = \binom{n+2}{2}$, and by MacMahon's formula $\text{PP}(2, 2, \ell) = (\ell + 1)(\ell + 2)^2(\ell + 3)/12$. If the empty word is not counted, H decreases by one and the inequalities above remain strict. For orientation: hypoplactic classes correspond to quasi-ribbon tableaux, which gives $h(n, \ell) = \sum_{k \geq 1} \binom{\ell-1}{k-1} \binom{n+\ell-k}{\ell}$ for $\ell \geq 1$, for example $h(2, \ell) = 2\ell$. For $n = 2$ this gives:

ℓ	0	1	2	3	4	5
$h(2, \ell)$	1	2	4	6	8	10
$H(2, \ell)$	1	3	7	13	21	31
$\text{PP}(2, 2, \ell)$	1	6	20	50	105	196

So $H(2, \ell)$ grows like ℓ^2 and $\text{PP}(2, 2, \ell)$ like $\ell^4/12$; the two sides agree only in the trivial case $\ell = 0$. This remark is context. It is not formalised and the disproof does not depend on it.

The earlier submission and its error

Pull request 280 of the competition repository (by `orionsheep`) proposed the instance $n = 2$, $\ell = 2$ with the values 6 and 20. It was closed without merging on 3 October 2026. The reviewer's stated reason was that the main Lean theorem was arithmetic with none of the objects of the conjecture: it asserted $2 + 4 = 6$, $6 \geq 6$, $17280/864 = 20$ and $20 > 6$; no hypoplactic class and no plane partition was defined in Lean; and the word count was entered by hand and omitted the empty word (there are 7 words of length at most 2, not 6).

That is the error. Its final Lean theorem `conjecture_refuted` is the conjunction of these four numerical facts. The statements “there are 6 words”, “the classes are at most as many as the words” and “the box contains 20 plane partitions” appear only in comments; the formal part says nothing about words, congruence classes or plane partitions, so it does not certify the counterexample. The mathematical idea was right: the classes of nonempty words of length at most 2 over two letters are at most 6 in number, the box contains 20 plane partitions, and so the instance $n = 2$, $\ell = 2$ refutes the identity, as the reviewer also noted.

The present submission defines the hypoplactic congruence on Mathlib’s free monoid, the sets of classes of given length, and plane partitions in a box; it proves the exact counts 4, 7 and 20, and negates the identity as a statement quantified over all n and ℓ .

Formal verification and scope

`KnuthRel` and `QuarticRel` are the four families of relations above on Mathlib’s `FreeMonoid` of a linearly ordered alphabet. `hypoCon` is the congruence generated by them (Mathlib’s `conGen`), and `Hypo` is the quotient monoid. `placticCon_le_hypoCon` states that Knuth equivalence is contained in it. The theorems `knuth_instance` and `quartic_instance` show $010 \equiv 100$ and $1010 \equiv 0101$ over two letters, so the relations are really present. Lemma 1 is

`length_eq_of_hypoCon, eq_or_three_le_of_hypoCon.`

The sets of elements of the hypoplactic monoid represented by a word of length exactly ℓ , respectively at most ℓ , are

`classesOfLength, classesUpToLength;`

their sizes are h and H . Part (a) is

`ncard_classesOfLength_of_le_two, ncard_classesUpToLength_two,`

and part (b) is

`ncard_classesOfLength_of_unique, ncard_classesUpToLength_of_unique.`

`PlanePartition a b c` is the type of arrays indexed by `Fin a` and `Fin b` with values in `Fin (c + 1)` that are weakly decreasing along rows and columns. The count 20 is `card_pp_222`, a kernel evaluation over all 81 arrays. The two lower bounds of (c) are

`card_pp_one_column, card_pp_height_one.`

The proposition `ClaimedIdentityUpTo` is the identity $H(n, \ell) = \text{PP}(2, n, \ell)$ for all n and ℓ , and `ClaimedIdentityExact` is the same with h . Their negations are

`conjecture_false_upTo, conjecture_false_exact.`

The theorems `two_letters_length_two`, `one_letter` and `length_one` are the three displayed consequences of Theorem 2. No `sorry`, `native_decide` or custom axiom occurs.

Not formalised: the exact binomial values and MacMahon’s formula, the symmetry of PP in its three arguments, the quasi-ribbon formula and the table in the remark, and the correspondence of the presentation used here with the definition of the hypoplactic monoid through quasi-ribbon tableaux. The counterexamples do not depend on the exact form of the relations: they use only that every defining relation preserves length and has length at least 3.

Source and reproduction

The exact bilingual source is included byte-for-byte as `SOURCE.md`. Its repository path is

`conjectures/00000003842.md.`

The project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b.`

From `lean/`, run `lake build`, then

`lake env lean -DwarningAsError=true Main.lean.`

The included public-Git manifest locks all dependency revisions. The `verification/` directory records the fresh-build logs, theorem-axiom reports, artifact hashes, review and final PDF inspection. No supplementary numerical computation is required.